Suggested Mark Scheme for 2017 JJC Prelim Paper 4 (9729/04)

1 (a) (i)

Titration number	1	2
final burette reading / cm^3	23.20	33.25
initial burette reading / cm^3	0.00	10.00
volume of FA2 (or $KMnO_4$) used / cm^3	23.20	23.25
	✓	✓

(ii) average volume of **FA2** (or $KMnO_4$) used = $\frac{23.20+23.25}{2}$ = 23.23 cm^3

(b) (i) mass of $KMnO_4$ in 23.23 cm^3 = 2.00 × $\frac{23.23}{1000}$ = 0.0465 g

amount of $KMnO_4$ used = $\frac{0.0465}{39.1+54.9+4(16.0)}$ = 2.93 × 10^{-4} mol

(ii) Since $5FeSO_4 \equiv 1KMnO_4$,

amount of $FeSO_4$ in 25.0 cm^3 of **FA 4** = 5 × (2.93 × 10^{-4}) = 1.47 × 10^{-3} mol

[$FeSO_4$] in **FA 4** = (1.47 × 10^{-3}) ÷ $\frac{25.0}{1000}$ = 0.0588 mol dm^{-3}

(iii) amount of $FeSO_4$ in 25.0 cm^3 of **FA 1** = amount of $FeSO_4$ in 250 cm^3 of **FA 4**

= 0.0588 × $\frac{250}{1000}$ = 0.0147 mol

[$FeSO_4$] in **FA 1** = 0.0147 ÷ $\frac{25.0}{1000}$ = 0.588 mol dm^{-3}

or

Using $c_1V_1 = c_2V_2$,

[$FeSO_4$] in FA 1 = 0.0588 × $\frac{250}{25.0}$ = 0.588 mol dm^{-3}

(iv) [$FeSO_4$] in **FA 1** in g dm^{-3} = 0.588 × [55.8 + 32.1 + 4(16.0)] = 89.3 g dm^{-3}

% by mass of $FeSO_4$ in **FA 1** = $\frac{89.3}{215.0}$ × 100 % = 41.5 %

%. error = $\frac{\text{error per reading} \times \text{no. of reading}}{\text{quantity measured}}$ × 100 %

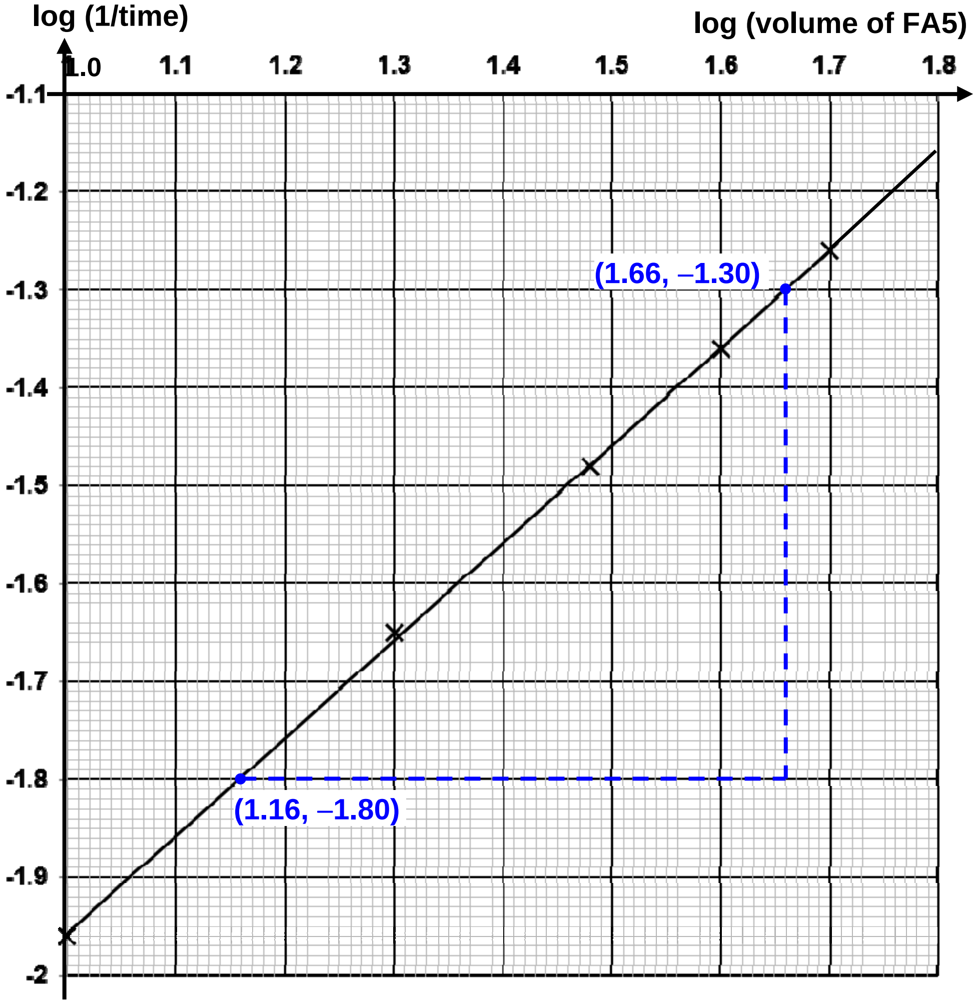
(c) (i) % error = $\frac{0.06\times 1}{25.0}$ × 100 % = 0.240 %

(ii) % error = $\frac{0.05\times 2}{23.23}$ × 100 % = 0.430 %

2 **(a)**

Experiment	volume of FA 5 / cm^3	volume of FA 3 / cm^3	volume of water / cm^3	t / s	log (1/*t*)	log (volume of FA 5)
1	50.0	5.0	0.0	18	–1.26	1.70
2	10.0	5.0	40.0	93	–1.96	1.00
3	20.0	5.0	30.0	45	–1.65	1.30
4	30.0	5.0	20.0	30	–1.48	1.48
5	40.0	5.0	10.0	23	–1.36	1.60

(b)



(c) gradient = $\frac{(-1.80)-(-1.30)}{1.16-1.66}$ = 1.00 = 1

The reaction is **first** order with respect to thiosulfate ions, $S_2O_3^{2-}$.

(d)

Experiment	Volume of **FA 5**/ cm^3	Volume of deionised water/ cm^3	Volume of **FA 3**/ cm^3
2	10.0	40.0	5.0
(i)	10.0	35.0	10.0
(ii)	10.0	25.0	20.0

2 **(e)** **(i)** Since $2H^+ \equiv 1H_2SO_4$ while $1H^+ \equiv 1HCl$, $[H^+]$ is halved. Since the reaction is first order w.r.t $[H^+]$, when $[H^+]$ is halved, the rate is halved and hence, the reaction time will be doubled.

(ii) Since the total volume of mixture is kept constant (*i.e.* 55 cm^3), the depth of the mixture in a 100 cm^3 beaker is greater and hence, the reaction time will be shorter.

3 **(a)** **(i)** On strong heating, pale green solid turns yellow (or brown) and becomes black eventually. Condensation (or steamy fumes) on test–tube.

Colourless, pungent gas turns damp red litmus blue.

Colourless, pungent gas turns damp blue litmus red.

Colourless, pungent gas decolourised purple $KMnO_4$.

3 **(a)** **(ii)**

test	*observations*	
	FA 6	**FA 7**
1. To a 0.5 cm depth of solution in a boiling tube add aqueous sodium hydroxide, then	Green ppt insoluble in excess NaOH(aq), turning brown on contact with air.	No ppt formed
Warm gently.	Colourless, pungent gas turns damp red litmus paper blue.	No gas evolved.
Allow to cool, add a piece of aluminium foil and warm again.		Effervescence observed. Colourless, pungent gas turns damp red litmus paper blue.
2. To a 1 cm depth of solution in at test-tube add a 1 cm depth of dilute sulfuric acid followed by a few drops of aqueous potassium manganate(VII).	Purple $KMnO_4$ decolourised.	Effervescence observed. Brown, pungent gas formed. Purple $KMnO_4$ decolourised.
3. To a 1 cm depth of solution in at test-tube add a 2 cm depth of hydrogen peroxide and leave to stand.	Effervescence observed. Colourless, odourless gas relights glowing splint. Solution turns yellow.	No gas evolved.
4. To a 1 cm depth of solution in at test-tube add a 1 cm depth of dilute hydrochloric acid, then	No gas evolved.	Effervescence observed. Brown, pungent gas formed.
Add a 1 cm depth of aqueous barium nitrate.	White ppt formed.	No ppt formed.

(b) **(i)**

	cation(s)	anion(s)
FA 6	Fe^{2+} , NH_4^+	SO_4^{2-}
FA 7	unknown	NO_2^-

3 **(b)** **(ii)** To 1 cm depth of the **FA 6** in a test–tube, add 1 cm depth of NH_3(aq).
Green ppt insoluble in excess NH_3(aq), turning brown on contact with air.

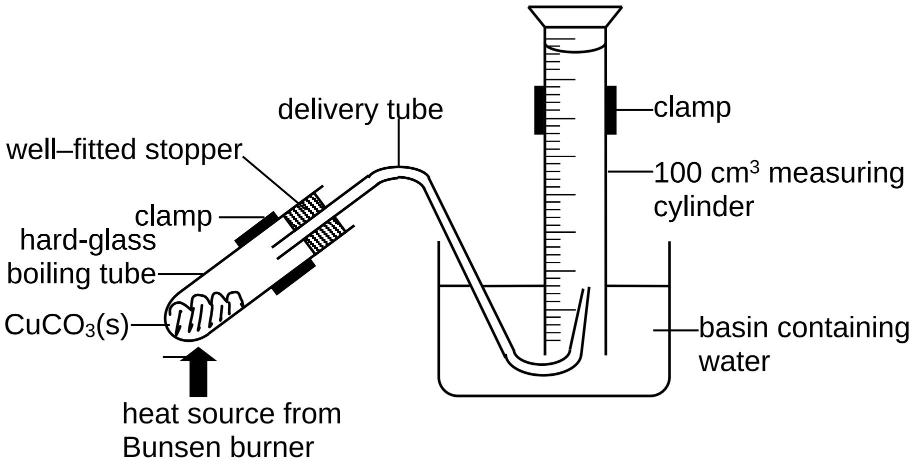
(iii) $Fe^{2+}(aq) + 2OH^-(aq) \rightarrow Fe(OH)_2(s)$
or
$[Fe(H_2O)_6]^{2+}(aq) + 2NH_3(aq) \rightarrow [Fe(OH)_2(H_2O)_4](s) + 2NH_4^+(aq)$

(c)

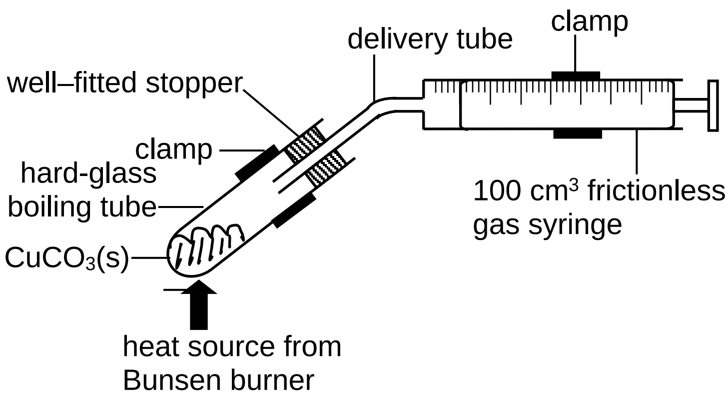
Test	Observation
To 1 cm depth of the 4 samples in separate test–tubes, add 1 cm depth of Tollens' reagent. Warm the mixture in hot water–bath.	For benzaldehyde, silver mirror is formed but not for the other 3 samples.
To 1 cm depth of the remaining 3 samples in separate test–tubes, add a few drops of acidified $KMnO_4$(aq) (or acidified $K_2Cr_2O_7$(aq)). Warm the mixture in hot water–bath.	For propan–2–ol, purple $KMnO_4$ decolourises (or orange $K_2Cr_2O_7$ turns green) but not for the other 2 samples.
To 1 cm depth of the remaining 2 samples in separate test–tubes, add 1 cm depth of alkaline I_2(aq). Warm the mixture in hot water–bath.	For propanone, pale yellow ppt of CHI_3 is formed but not for the other sample.
To 1 cm depth of the last sample in a test–tube, add 1 small spatula of PCl_5(s).	For 2–methylpropan–2–ol, white fumes of HC*l* observed.

*The last two steps can be swopped.

4 **(a)** Downward displacement gas collection method:



Gas syringe method:



(b) If a 100 cm^3 capacity apparatus is used,

maximum amount of gas evolved = $\frac{100}{1000} \div 25.3 = 3.95 \times 10^{-3}$ mol

For Equation 4.1, since $2CuCO_3 \equiv 2.5$ gas ($CO_2 + O_2$),

amount of $CuCO_3$ required = $\frac{2}{2.5} \times (3.95 \times 10^{-3}) = 3.16 \times 10^{-3}$ mol

mass of $CuCO_3$ required = $(3.16 \times 10^{-3}) \times 123.5 = 0.391$ g

For Equation 4.2, since $1CuCO_3 \equiv 1\ CO_2$,
mass of $CuCO_3$ required = $(3.95 \times 10^{-3}) \times 123.5 = 0.488$ g

Experimental Procedures:

step 1: Using an electronic weighing balance precise to 3 decimal places, weigh a clean and dry boiling tube and record its mass as m_1.

step 2: Weigh accurately 0.391 g of $CuCO_3$ into the boiling tube and record the total mass as m_2.

$\therefore$ mass of $CuCO_3$ used = $(m_2 - m_1)$ g

step 3: Set up the apparatus as shown in part (a).

step 4: Recording the initial syringe (or burette or measuring cylinder) reading as V_1.

step 5: Light up the Bunsen burner and heat the tube with its content gently then strongly. Decomposition is complete when the syringe piston stops moving (or water level in measuring cylinder/burette stops falling).

step 6: Recording the final syringe (or burette or measuring cylinder) reading as V_2.

$\therefore$ volume of gas collected = $(V_2 - V_1)$ cm^3

amount of $CuCO_3$ used = $\frac{m_2 - m_1}{123.5} = x$ mol

amount of gas collected = $\frac{V_2 - V_1}{1000} \div 25.3 = y$ mol

Then determine the ratio of $x : y$. If the ratio is about 2 : 2.5 (or 4 : 5), Equation 4.1 represent the actual decomposition of $CuCO_3$. If the ratio is about 1 : 1, Equation 4.2 represent the actual decomposition of $CuCO_3$.